Math 1610, Honors Calculus, Fall 2026, practice problems for Exam 2 — Solutions

Please note. These solutions were generated agentically (by an AI assistant). They have been looked over, but they may still contain errors. If you find a mistake, or anything that is unclear or could be better explained, please let us know — we would be glad to hear about it.

Conventions and facts we use.

Limits and continuity. As in lecture, $\lim_{x\to c} f(x) = L$ means: for every $\epsilon > 0$ there is a $\delta > 0$ such that $|f(x) - L| < \epsilon$ whenever $0 < |x - c| < \delta$. A function is continuous at c if $\lim_{x\to c} f(x) = f(c)$. For a vector-valued $\vec{F}$, the definition is the same with the absolute value replaced by the length: $\|\vec{F}(t) - \vec{L}\| < \epsilon$ whenever $0 < |t - c| < \delta$.

Two values can't both be close to one number. Several times below we show that a limit does not exist in the same way. Suppose a quantity takes two values p and q with $|p - q| \geq 2\epsilon$. Then no single number L is within ϵ of both, because if it were, then

$$|p - q| \leq |p - L| + |L - q| < \epsilon + \epsilon = 2\epsilon.$$

The same holds for vectors, using the triangle inequality for lengths.

The graphing rules (Lecture 11, from the mean value theorem). If f is continuous on $[a, b]$ and $f'(x) > 0$ for all $x \in (a, b)$, then f is strictly increasing on $[a, b]$; if $f'(x) < 0$ throughout, f is strictly decreasing. A strictly increasing (or decreasing) function never takes the same value twice.

1. Show, directly from the definition of the limit, that $\lim\limits_{x\to 2}\big(x^2 + x\big) = 6$.

Solution: Scratch work. We need to make $|x^2 + x - 6|$ small. It factors:

$$|x^2 + x - 6| = |x - 2|\,|x + 3|.$$

The first factor is the one we control. The second factor is not small, but it is not large either, as long as x stays near 2. If we insist that $|x-2| < 1$, then $1 < x < 3$, so $4 < x+3 < 6$ and $|x+3| < 6$. Then $|x^2 + x - 6| < 6\,|x - 2|$, and this is less than ϵ once $|x - 2| < \epsilon/6$. So we want δ to be at most 1 and at most $\epsilon/6$.

Proof. Let $\epsilon > 0$, and set $\delta = \min\{1, \epsilon/6\}$, which is positive. Suppose $0 < |x - 2| < \delta$. Since $\delta \leq 1$ we have $1 < x < 3$, so $|x + 3| < 6$. Since $\delta \leq \epsilon/6$ we have $|x - 2| < \epsilon/6$. Therefore

$$|(x^2 + x) - 6| = |x - 2|\,|x + 3| < \frac{\epsilon}{6} \cdot 6 = \epsilon.$$

This is exactly what the definition asks for, so $\lim_{x\to 2}(x^2 + x) = 6$.

2. Prove the sign preservation lemma: if g is continuous at c and $g(c) > 0$, then there is an $h > 0$ with $g(x) > 0$ for every $x \in (c-h, c+h)$.

Solution: (As in lecture, g is defined on an open interval around c, and we take h small enough that $(c-h, c+h)$ lies inside it.)

The idea: $g(x)$ is close to $g(c)$ when x is close to c, and if "close" means "closer than $g(c)$ itself", then $g(x)$ can't reach 0.

Continuity at c says that for every $\epsilon > 0$ there is a $\delta > 0$ with $|g(x) - g(c)| < \epsilon$ whenever $|x - c| < \delta$. (At $x = c$ itself this is automatic, since $|g(c) - g(c)| = 0$.) Apply this with the particular choice $\epsilon = g(c)$, which is allowed because $g(c) > 0$. We get a $\delta > 0$ such that, whenever $|x - c| < \delta$,

$$-g(c) < g(x) - g(c) < g(c).$$

Adding $g(c)$ to the left-hand inequality gives $0 < g(x)$. So $h = \delta$ works: every $x \in (c-h, c+h)$ has $g(x) > 0$.

3. Let $f : [a, b] \to \mathbb{R}$ be continuous, with $f(x) > 0$ for every $x \in [a, b]$.

   (a) Show that there is a number $m > 0$ with $f(x) \geq m$ for every $x \in [a, b]$.

   Solution: Since f is continuous on the closed bounded interval $[a, b]$, the extreme value theorem says that f attains a minimum value: there is a point $x_0 \in [a, b]$ with $f(x_0) \leq f(x)$ for every $x \in [a, b]$. Let $m = f(x_0)$. Then $f(x) \geq m$ for every $x \in [a, b]$, and $m > 0$ because x_0 is one of the points of $[a, b]$, where f is positive.

   It is worth noticing what the extreme value theorem bought us. Knowing that each value $f(x)$ is positive does not, by itself, stop the values from getting closer and closer to 0. What stops that is that the smallest value is actually attained, at a point where f is positive.

   (b) Give an example showing that this can fail if $[a, b]$ is replaced by the interval $[0, \infty)$.

   Solution: Take $f(x) = \dfrac{1}{1+x}$ on $[0, \infty)$. The denominator is at least 1, so f is defined and positive everywhere on $[0, \infty)$, and it is continuous there, being a quotient of polynomials with a denominator that never vanishes.

   Suppose there were an $m > 0$ with $f(x) \geq m$ for every $x \geq 0$. Pick any $x > \frac{1}{m}$ (for example $x = \frac{2}{m}$). Then $1 + x > x > \frac{1}{m}$, so

   $$f(x) = \frac{1}{1+x} < m,$$

   a contradiction. So no such m exists.

   The hypothesis that fails is that the interval is bounded: the values of f creep down toward 0 as x goes off to infinity, and there is no point where the smallest value is attained.

4. Let $f : [0,2] \to \mathbb{R}$ be continuous, with $f(0) = f(2)$. Show that there is some $x \in [0,1]$ with $f(x) = f(x+1)$.

Solution: Define $g(x) = f(x+1) - f(x)$ for $x \in [0,1]$. This makes sense, because for $x \in [0,1]$ both x and $x+1$ lie in $[0,2]$. We want a point where $g(x) = 0$.

g is continuous on $[0,1]$. The function $x \mapsto f(x)$ is continuous by assumption. The function $x \mapsto f(x+1)$ is also continuous, and we can check this directly. Fix $c \in [0,1]$ and $\epsilon > 0$. Since f is continuous at $c+1$, there is a $\delta > 0$ with $|f(y) - f(c+1)| < \epsilon$ whenever $y \in [0,2]$ and $|y - (c+1)| < \delta$. Now if $|x - c| < \delta$, then $y = x + 1$ satisfies $|y - (c+1)| = |x - c| < \delta$, so $|f(x+1) - f(c+1)| < \epsilon$. Shifting the input by 1 just shifts the picture sideways, and continuity survives. A difference of continuous functions is continuous, so g is continuous.

Comparing the endpoints. We have

$$g(0) = f(1) - f(0), \qquad g(1) = f(2) - f(1) = f(0) - f(1) = -g(0),$$

using $f(2) = f(0)$. There are two cases.

- If $g(0) = 0$, then $f(1) = f(0)$, and $x = 0$ works.
- If $g(0) \neq 0$, then $g(0)$ and $g(1) = -g(0)$ are nonzero with opposite signs, so 0 lies between them. By the intermediate value theorem there is an $x \in [0,1]$ with $g(x) = 0$, that is, $f(x+1) = f(x)$.

A picture. If you walk along the graph of f from height $f(0)$ and come back to the same height at $x = 2$, then somewhere there is a horizontal chord of length exactly 1 joining two points of the graph.

5. (Negating uniform continuity.)

   (a) Write out, without using the word "not", what it means for $f : S \to \mathbb{R}$ to fail to be uniformly continuous on S.

   Solution: Recall the definition: f is uniformly continuous on S if

   > for every $\epsilon > 0$ there is a $\delta > 0$ such that for all $x_1, x_2 \in S$ with $|x_1 - x_2| < \delta$, we have $|f(x_1) - f(x_2)| < \epsilon$.

   To negate it, go through the statement from left to right, turning each "for every" into "there is" and each "there is" into "for every", and negate the final inequality. The result: f fails to be uniformly continuous on S if

   > there is an $\epsilon > 0$ such that for every $\delta > 0$ there are points $x_1, x_2 \in S$ with $|x_1 - x_2| < \delta$ and $|f(x_1) - f(x_2)| \geq \epsilon$.

   In words: there is a fixed gap ϵ such that, no matter how close together you demand the inputs be, you can find two inputs that close whose outputs are at least ϵ apart.

(b) Show that $f(x) = 1/x$ is not uniformly continuous on the open interval $(0, 1)$.

Solution: We verify the statement from part (a) with $\epsilon = 1$.
Let $\delta > 0$ be given. Choose a whole number $n \geq 2$ that is also large enough that $\frac{1}{2n} < \delta$ (possible by the Archimedean property: take $n > \frac{1}{2\delta}$ as well as $n \geq 2$). Set

$$x_1 = \frac{1}{n}, \qquad x_2 = \frac{1}{2n}.$$

Both lie in $(0, 1)$, since $n \geq 2$. They are close together:

$$|x_1 - x_2| = \frac{1}{n} - \frac{1}{2n} = \frac{1}{2n} < \delta.$$

But their outputs are far apart:

$$|f(x_1) - f(x_2)| = |n - 2n| = n \geq 1 = \epsilon.$$

Since δ was arbitrary, f is not uniformly continuous on $(0, 1)$.

What goes wrong is near 0: the graph of $1/x$ gets steeper and steeper, so a fixed δ that works near $x = \frac{1}{2}$ is far too large near $x = 0.001$. Notice that f is still continuous at every point of $(0, 1)$. The theorem from lecture (continuous on $[a, b]$ implies uniformly continuous) doesn't apply, because $(0, 1)$ is not closed, and $1/x$ can't be extended continuously to 0.

6. Let $f(x) = x\,|x|$.

(a) Show, from the definition, that f is differentiable at 0 with $f'(0) = 0$.

Solution: For $h \neq 0$, the difference quotient at 0 is

$$\frac{f(0 + h) - f(0)}{h} = \frac{h\,|h| - 0}{h} = |h|.$$

So we need $\lim_{h \to 0} |h| = 0$. Given $\epsilon > 0$, take $\delta = \epsilon$: if $0 < |h| < \delta$ then $\big|\, |h| - 0 \,\big| = |h| < \epsilon$. Hence f is differentiable at 0 and $f'(0) = 0$.

(b) Compute $f'(x)$ for $x \neq 0$, and write down the function f' on all of $\mathbb{R}$.

Solution: Write the function in pieces:

$$f(x) = \begin{cases} x^2 & x \geq 0, \\ -x^2 & x < 0. \end{cases}$$

At a point $x > 0$. The derivative at x depends only on the values of f near x, since the difference quotient $\frac{f(x+h)-f(x)}{h}$ only uses f at x and $x + h$, and in the limit only small h matter. If $|h| < x$, then $x + h > 0$, so on that range of h the difference quotient of f is exactly the difference quotient of x^2. Therefore $f'(x)$ exists and equals the derivative of x^2, which is $2x$.
At a point $x < 0$. In the same way, for $|h| < |x|$ the point $x + h$ is negative, f agrees with $-x^2$ there, and $f'(x) = -2x$.

All together. Combining these with part (a),

$$f'(x) = \begin{cases} 2x & x > 0, \\ 0 & x = 0, \\ -2x & x < 0 \end{cases} \qquad \text{that is,} \qquad f'(x) = 2|x| \text{ for every } x.$$

(c) Is f' differentiable at 0?

Solution: No. By part (b), $f'(x) = 2|x|$, so the difference quotient of f' at 0 is

$$\frac{f'(h) - f'(0)}{h} = \frac{2|h|}{h} = \begin{cases} 2 & h > 0, \\ -2 & h < 0. \end{cases}$$

Suppose this had a limit L as $h \to 0$. Take $\epsilon = 2$ in the definition of the limit. Then there would be a $\delta > 0$ such that the quotient is within 2 of L for every h with $0 < |h| < \delta$. Both $h = \delta/2$ and $h = -\delta/2$ are such values of h, so both 2 and -2 would be within 2 of L. They are 4 apart, so this is impossible (see "Two values can't both be close to one number" above). So the limit does not exist, and f' is not differentiable at 0.

So $f(x) = x|x|$ is differentiable everywhere, and its derivative is continuous, but the derivative has a corner at 0, just like $|x|$. Differentiating once smooths out the corner of $|x|$ by one step, but not by two.

7. (Symmetric difference quotients.)

(a) Suppose f is differentiable at c. Show that $\lim_{h \to 0} \dfrac{f(c+h) - f(c-h)}{2h} = f'(c)$.

Solution: The idea is to split the symmetric quotient into two ordinary difference quotients, one looking to the right of c and one looking to the left. Add and subtract $f(c)$ in the numerator:

$$\frac{f(c+h) - f(c-h)}{2h} = \frac{1}{2} \cdot \frac{f(c+h) - f(c)}{h} + \frac{1}{2} \cdot \frac{f(c-h) - f(c)}{-h}.$$

(For the second term, note that $-(f(c-h) - f(c))/h$ is the same as $(f(c-h) - f(c))/(-h)$.)
The first term. By the definition of $f'(c)$, $\frac{f(c+h)-f(c)}{h} \to f'(c)$ as $h \to 0$.
The second term. This is the ordinary difference quotient with $-h$ in place of h. We check that it also tends to $f'(c)$. Let $\epsilon > 0$. From the definition of $f'(c)$ there is a $\delta > 0$ such that

$$\left| \frac{f(c+k) - f(c)}{k} - f'(c) \right| < \epsilon \quad \text{whenever } 0 < |k| < \delta.$$

If $0 < |h| < \delta$, then $k = -h$ also satisfies $0 < |k| < \delta$, so $\left| \frac{f(c-h)-f(c)}{-h} - f'(c) \right| < \epsilon$. That is exactly the statement that the second term tends to $f'(c)$ as $h \to 0$.
Putting them together. Both terms have limits, so by the limit laws (a constant times a limit, and the sum of two limits),

$$\lim_{h \to 0} \frac{f(c+h) - f(c-h)}{2h} = \frac{1}{2} f'(c) + \frac{1}{2} f'(c) = f'(c).$$

(b) Show that for $f(x) = |x|$ the limit in part (a) exists at $c = 0$, even though f is not differentiable there. So the converse of (a) is false.

Solution: The symmetric limit exists. For $h \neq 0$,

$$\frac{f(0+h) - f(0-h)}{2h} = \frac{|h| - |-h|}{2h} = \frac{0}{2h} = 0,$$

since $|-h| = |h|$. A quantity that equals 0 for every $h \neq 0$ has limit 0, so the symmetric limit exists (and equals 0).

f is not differentiable at 0. The ordinary difference quotient is

$$\frac{f(h) - f(0)}{h} = \frac{|h|}{h} = \begin{cases} 1 & h > 0, \\ -1 & h < 0. \end{cases}$$

Exactly as in Problem 6(c) (now with $\epsilon = 1$, and values 1 and -1 that are 2 apart), no number L can be within 1 of both values, and both values occur for h as close to 0 as we like. So the limit does not exist.

What happened: the symmetric quotient averages the slope on the right $(+1)$ with the slope on the left (-1), and the average is 0. The symmetric quotient can't see a corner, because the corner is symmetric. So knowing the symmetric limit exists does not tell you that f is differentiable.

8. Use the mean value theorem to show that for $0 < a < b$,

$$3a^2(b-a) \;<\; b^3 - a^3 \;<\; 3b^2(b-a).$$

Solution: Let $f(x) = x^3$. It is a polynomial, so it is continuous on $[a, b]$ and differentiable on (a, b), with $f'(x) = 3x^2$. The mean value theorem gives a point $\xi \in (a, b)$ with

$$b^3 - a^3 = f(b) - f(a) = f'(\xi)(b-a) = 3\xi^2(b-a).$$

Now we use where ξ is. Since $0 < a < \xi < b$, squaring preserves the inequalities (for positive numbers, $x < y$ gives $x^2 < y^2$; this is the fact from Lecture 8 that x^n is increasing on the positive numbers), so

$$a^2 < \xi^2 < b^2.$$

Multiply through by the positive number $3(b-a)$:

$$3a^2(b-a) < 3\xi^2(b-a) < 3b^2(b-a).$$

The middle term is $b^3 - a^3$, which is what we wanted.

In words: the average slope of x^3 over $[a, b]$ lies strictly between the smallest slope ($3a^2$, at the left end) and the largest slope ($3b^2$, at the right end).

9. Let f be continuous on $[a,b]$ and twice differentiable on (a,b), with $f(a) = f(b) = 0$, and suppose $f(c) > 0$ for some $c \in (a,b)$.

   (a) Use the mean value theorem on $[a,c]$ and on $[c,b]$ to find points $\xi_1 < \xi_2$ with $f'(\xi_1) > 0$ and $f'(\xi_2) < 0$.

Solution: First, the mean value theorem applies on both smaller intervals: f is continuous on $[a,b]$, hence on $[a,c]$ and on $[c,b]$, and it is differentiable on (a,b), hence on (a,c) and on (c,b).
On $[a,c]$. There is a $\xi_1 \in (a,c)$ with

$$f'(\xi_1) = \frac{f(c) - f(a)}{c - a} = \frac{f(c)}{c - a} > 0,$$

since $f(c) > 0$ and $c - a > 0$.
On $[c,b]$. There is a $\xi_2 \in (c,b)$ with

$$f'(\xi_2) = \frac{f(b) - f(c)}{b - c} = \frac{-f(c)}{b - c} < 0.$$

Finally $\xi_1 < c < \xi_2$, so $\xi_1 < \xi_2$. (The graph has to climb from 0 up to $f(c)$, and then come back down to 0.)

   (b) Show that $f''(\xi) < 0$ for some $\xi \in (a,b)$.

Solution: Now apply the mean value theorem a second time, to the function f' on the interval $[\xi_1, \xi_2]$.
Check the hypotheses. We are told f is twice differentiable on (a,b), which means f' is differentiable at every point of (a,b). In particular f' is differentiable on (ξ_1, ξ_2), and, since a differentiable function is continuous, f' is continuous on $[\xi_1, \xi_2]$ (this interval lies inside (a,b)).
So there is a $\xi \in (\xi_1, \xi_2)$ with

$$f''(\xi) = \frac{f'(\xi_2) - f'(\xi_1)}{\xi_2 - \xi_1}.$$

The numerator is (negative) − (positive), which is negative, and the denominator is positive. So $f''(\xi) < 0$, and ξ lies in $(\xi_1, \xi_2) \subseteq (a,b)$.

The picture: a graph that starts at 0, rises above 0, and returns to 0 must bend downward somewhere.

10. (The second derivative test, without continuity of f''.) Let f be differentiable on an open interval (a,b), let $c \in (a,b)$ with $f'(c) = 0$, and suppose that $f''(c)$ exists and $f''(c) > 0$.

   (a) Define $Q(x) = \dfrac{f'(x) - f'(c)}{x - c}$ for $x \neq c$, and $Q(c) = f''(c)$. Show that Q is continuous at c.

Solution: The number $f''(c)$ is, by definition, the derivative of the function f' at c:

$$f''(c) = \lim_{x \to c} \frac{f'(x) - f'(c)}{x - c}.$$

(Here we've written the difference quotient with x in place of $c + h$. This is the same limit, with $x = c + h$.)

For $x \neq c$, the expression inside this limit is exactly $Q(x)$. The limit only looks at $x \neq c$, so this says

$$\lim_{x \to c} Q(x) = f''(c) = Q(c).$$

That is precisely the definition of Q being continuous at c. We defined $Q(c)$ to be the value that fills in the gap.

(b) Show that there is an $h > 0$ such that $f'(x) < 0$ for $x \in (c-h, c)$ and $f'(x) > 0$ for $x \in (c, c+h)$.

Solution: By part (a), Q is continuous at c, and $Q(c) = f''(c) > 0$. By sign preservation there is an $h > 0$ with $Q(x) > 0$ for every $x \in (c-h, c+h)$. Shrinking h if necessary, we may also assume $(c-h, c+h) \subseteq (a, b)$.

Since $f'(c) = 0$, for $x \neq c$ we have $Q(x) = \dfrac{f'(x)}{x-c}$, so

$$f'(x) = Q(x)\,(x-c).$$

- If $x \in (c, c+h)$, then $Q(x) > 0$ and $x - c > 0$, so $f'(x) > 0$.
- If $x \in (c-h, c)$, then $Q(x) > 0$ and $x - c < 0$, so $f'(x) < 0$.

Another way, without Q (this is how we did it in the review session). Since $f'(c) = 0$, the definition of $f''(c)$ says

$$\lim_{k \to 0} \frac{f'(c+k)}{k} = f''(c) > 0.$$

Use the definition of this limit with $\epsilon = f''(c)$. There is a $\delta > 0$ such that, whenever $0 < |k| < \delta$,

$$\left| \frac{f'(c+k)}{k} - f''(c) \right| < f''(c), \qquad \text{and so} \qquad \frac{f'(c+k)}{k} > 0.$$

A fraction is positive when the top and bottom have the same sign. So $f'(c+k) > 0$ when $0 < k < \delta$, and $f'(c+k) < 0$ when $-\delta < k < 0$. Taking $h = \delta$ (made smaller if needed so that $(c-h, c+h)$ stays inside (a, b)) gives part (b). This is really the proof of sign preservation, carried out by hand for this one quotient.

(c) Conclude that f has a local minimum at c.

Solution: Let h be as in part (b). We show $f(x) \geq f(c)$ for every $x \in (c-h, c+h)$. (Note that f is differentiable on (a, b), hence continuous there, so the mean value theorem can be applied on any closed interval inside $(c-h, c+h)$.)

- If $x \in (c, c+h)$: the mean value theorem on $[c, x]$ gives a $\xi \in (c, x)$ with $f(x) - f(c) = f'(\xi)(x-c)$. Here $f'(\xi) > 0$ by part (b), and $x - c > 0$, so $f(x) > f(c)$.
- If $x \in (c-h, c)$: the mean value theorem on $[x, c]$ gives a $\xi \in (x, c)$ with $f(c) - f(x) = f'(\xi)(c-x)$. Here $f'(\xi) < 0$ by part (b), and $c - x > 0$, so $f(c) - f(x) < 0$, that is, $f(x) > f(c)$.
- If $x = c$, then $f(x) = f(c)$.

So $f(c) \leq f(x)$ on $(c-h, c+h)$, and c is a local minimum. (This is the first derivative test from lecture: f' changes from negative to positive at c.)

Why bother with Q? The proof in lecture applied sign preservation to f'' itself, which needs f'' to be continuous at c. Here f'' might not even exist anywhere except at c. The function Q is a way to package the single number $f''(c)$ as the value of a function that is continuous at c, so that sign preservation can be used anyway. This is the same trick as in the proof of the chain rule.

11. Let $f(x) = x^3 - 3x$.

(a) Find the critical points of f, and use the second derivative test to decide whether each is a local maximum or a local minimum.

Solution: The derivative is

$$f'(x) = 3x^2 - 3 = 3(x-1)(x+1),$$

which is 0 exactly when $x = 1$ or $x = -1$. These are the critical points.
The second derivative is $f''(x) = 6x$, which is continuous (so the version of the test proved in lecture applies).

- $f''(-1) = -6 < 0$, so $x = -1$ is a local maximum, with $f(-1) = -1 + 3 = 2$.
- $f''(1) = 6 > 0$, so $x = 1$ is a local minimum, with $f(1) = 1 - 3 = -2$.

(b) Explain why f attains a largest and a smallest value on $[-2, 3]$, and why each must occur either at an endpoint or at a critical point. Find both values, and every point where they are attained.

Solution: They exist. f is a polynomial, so it is continuous on the closed bounded interval $[-2, 3]$. By the extreme value theorem it attains a largest and a smallest value there.
Where they can be. Suppose the largest value is attained at a point x_0 that is not an endpoint, so $-2 < x_0 < 3$. Then $f(x_0) \geq f(x)$ for all x in $[-2, 3]$, and in particular for all x in a small open interval around x_0. So x_0 is a local maximum. Since f is differentiable at x_0, the theorem from lecture (an interior local extremum of a differentiable function has $f' = 0$) gives $f'(x_0) = 0$, so x_0 is a critical point. The same argument works for the smallest value. So we only need to compare the values at the endpoints $-2, 3$ and at the critical points $-1, 1$ (both of which lie in $[-2, 3]$):

x	-2	-1	1	3
$f(x)$	$-8 + 6 = -2$	2	-2	$27 - 9 = 18$

Answer. The largest value is 18, attained only at $x = 3$. The smallest value is -2, attained at two points, $x = -2$ and $x = 1$. (No other point of $[-2, 3]$ can give these values: any point where an extreme value is attained is in our list of four.)
Notice that the local maximum at -1 is not the largest value on $[-2, 3]$: a local maximum only beats its neighbors.

(c) How many solutions does the equation $x^3 - 3x = 1$ have? Prove your answer.

Solution: Answer: exactly three. Let $g(x) = x^3 - 3x - 1$; we count the zeros of g. Note $g' = f'$, so g has the same critical points ± 1.
At least three. Some values:

$$g(-2) = -3, \qquad g(-1) = 1, \qquad g(1) = -3, \qquad g(2) = 1.$$

The sign changes from -2 to -1, from -1 to 1, and from 1 to 2. Since g is continuous, the intermediate value theorem gives a zero in each of $(-2, -1)$, $(-1, 1)$ and $(1, 2)$. These intervals don't overlap, so these are three different zeros.
At most three. Split the real line into three pieces: $(-\infty, -1]$, $[-1, 1]$ and $[1, \infty)$.

- On $(-\infty, -1]$: for $x < -1$ we have $g'(x) = 3(x-1)(x+1) > 0$ (both factors negative). If $x_1 < x_2 \leq -1$, the graphing rules on $[x_1, x_2]$ give $g(x_1) < g(x_2)$. So g is strictly increasing on this piece.
- On $[-1, 1]$: for $-1 < x < 1$, $g'(x) < 0$ (one factor negative, one positive), so g is strictly decreasing on $[-1, 1]$.
- On $[1, \infty)$: for $x > 1$, $g'(x) > 0$, so as in the first piece g is strictly increasing there.

A strictly increasing or strictly decreasing function takes the value 0 at most once. So each of the three pieces contains at most one zero of g. The pieces cover the whole real line, so g has at most three zeros. (A zero at a shared endpoint ± 1 would be counted twice, but this doesn't happen: $g(-1) = 1$ and $g(1) = -3$.)
Together: exactly three solutions.

12. (The fundamental theorem.) In parts (b) and (c), f is continuous on $\mathbb{R}$.

(a) Using the evaluation rule, compute $\displaystyle\int_1^2 \left(3x^2 - \frac{2}{x^2}\right) dx$, and check that the antiderivative you use really has the integrand as its derivative.

Solution: Finding an antiderivative. We run the derivative rules backwards. The derivative of x^3 is $3x^2$. The derivative of $\frac{1}{x}$ is $-\frac{1}{x^2}$ (the reciprocal rule from lecture), so the derivative of $\frac{2}{x}$ is $-\frac{2}{x^2}$. So try

$$P(x) = x^3 + \frac{2}{x}.$$

Checking. For $x \neq 0$, by the sum and scalar rules,

$$P'(x) = 3x^2 + 2 \cdot \left(-\frac{1}{x^2}\right) = 3x^2 - \frac{2}{x^2}.$$

This holds on the whole interval $[1, 2]$, where $x \neq 0$, and the integrand is continuous there. So P is an antiderivative of the integrand on $[1, 2]$.
Evaluating. By the evaluation rule,

$$\int_1^2 \left(3x^2 - \frac{2}{x^2}\right) dx = P(2) - P(1) = \Big(8 + 1\Big) - \Big(1 + 2\Big) = 6.$$

(b) Find a continuous f with $\int_0^x f(t)\,dt = x^3 + x$ for every x, and explain why it is the only one.

Solution: It is the only possible one. Suppose f is continuous and $\int_0^x f(t)\,dt = x^3 + x$ for every x. The left-hand side, as a function of x, is $F(x) = \int_0^x f(t)\,dt$. Because f is continuous, the fundamental theorem says F is differentiable with $F'(x) = f(x)$ at every x. But $F(x) = x^3 + x$, whose derivative is $3x^2 + 1$. So

$$f(x) = 3x^2 + 1 \qquad \text{for every } x.$$

Any continuous f that works must be this one.
It does work. $f(x) = 3x^2 + 1$ is continuous, and $P(x) = x^3 + x$ is an antiderivative of it. By the evaluation rule, $\int_0^x (3t^2 + 1)\,dt = P(x) - P(0) = x^3 + x$ for every x.
So $f(x) = 3x^2 + 1$, and it is the only one.

(c) Is there a continuous f with $\int_0^x f(t)\,dt = x^3 + 1$ for every x?

Solution: No. Put $x = 0$. The left-hand side is $\int_0^0 f(t)\,dt = 0$, the integral over an interval of length zero, whatever f is. The right-hand side is $0^3 + 1 = 1$. Since $0 \neq 1$, no function at all (continuous or not) can satisfy the equation for every x.

This is worth comparing with part (b). If you just differentiate, you get $f(x) = 3x^2$, which looks like an answer. But differentiating loses information about constants: $x^3 + 1$ and x^3 have the same derivative, and only one of them is 0 at $x = 0$. A function of the form $\int_0^x f$ always starts at 0 when $x = 0$.

13. (Cauchy–Schwarz at work.)

(a) Show that for all positive real numbers a, b, c, $(a + b + c)\Big(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\Big) \geq 9$.

Solution: Choosing the vectors. Cauchy–Schwarz compares $(\vec{u} \cdot \vec{v})^2$ with $\|\vec{u}\|^2 \, \|\vec{v}\|^2$. The left-hand side of our inequality is a product of two sums, which looks like $\|\vec{u}\|^2 \|\vec{v}\|^2$ if

$$\|\vec{u}\|^2 = a + b + c, \qquad \|\vec{v}\|^2 = \frac{1}{a} + \frac{1}{b} + \frac{1}{c}.$$

That suggests taking square roots of the entries. Since a, b, c are positive, they have positive square roots, and we set

$$\vec{u} = \big(\sqrt{a},\ \sqrt{b},\ \sqrt{c}\big), \qquad \vec{v} = \Big(\frac{1}{\sqrt{a}},\ \frac{1}{\sqrt{b}},\ \frac{1}{\sqrt{c}}\Big).$$

Then $\|\vec{u}\|^2 = a + b + c$ and $\|\vec{v}\|^2 = \frac{1}{a} + \frac{1}{b} + \frac{1}{c}$, as we wanted, and the dot product is very simple:

$$\vec{u} \cdot \vec{v} = \sqrt{a} \cdot \frac{1}{\sqrt{a}} + \sqrt{b} \cdot \frac{1}{\sqrt{b}} + \sqrt{c} \cdot \frac{1}{\sqrt{c}} = 1 + 1 + 1 = 3.$$

Applying Cauchy–Schwarz. We get $3 = |\vec{u} \cdot \vec{v}| \leq \|\vec{u}\| \, \|\vec{v}\|$. Both sides are nonnegative, so we may square, which gives

$$9 \leq \|\vec{u}\|^2 \, \|\vec{v}\|^2 = (a + b + c)\Big(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\Big).$$

(b) When does equality hold?

Solution: Answer: exactly when $a = b = c$.
Equality in our inequality means $9 = \|\vec{u}\|^2\|\vec{v}\|^2$. Taking square roots, it is the same as $3 = \|\vec{u}\|\,\|\vec{v}\|$, that is, equality in Cauchy–Schwarz for $\vec{u}$ and $\vec{v}$. By the equality case of Cauchy–Schwarz, this happens exactly when one of $\vec{u}, \vec{v}$ is a scalar multiple of the other.
Neither vector is zero (all their entries are positive). Suppose $\vec{u} = \lambda\vec{v}$ for some number λ. Comparing first entries, $\sqrt{a} = \lambda/\sqrt{a}$, so $a = \lambda$. Comparing the other entries in the same way, $b = \lambda$ and $c = \lambda$. So $a = b = c$. (If instead $\vec{v} = \mu\vec{u}$, the same comparison gives $a = b = c = 1/\mu$.)
Conversely, if $a = b = c$, then the left-hand side is $3a \cdot \frac{3}{a} = 9$, so equality holds.

14. (Vector-valued functions.)

(a) Show, directly from the definition, that $\lim_{t\to 0}(t,\, t^2) = (0, 0)$.

Solution: Scratch work. We need to make the length

$$\|(t, t^2) - (0, 0)\| = \sqrt{t^2 + t^4}$$

small. If $|t| < 1$, then $t^4 = t^2 \cdot t^2 \le t^2$, so $t^2 + t^4 \le 2t^2$, and the length is at most $\sqrt{2t^2} = \sqrt{2}\,|t| < 2|t|$. That is less than ϵ once $|t| < \epsilon/2$.

Proof. Let $\epsilon > 0$ and set $\delta = \min\{1, \epsilon/2\}$. Suppose $0 < |t| < \delta$. Since $|t| < 1$, we have $t^2 < 1$, so $t^4 \le t^2$ and

$$\|(t, t^2) - (0, 0)\| = \sqrt{t^2 + t^4} \le \sqrt{2t^2} = \sqrt{2}\,|t| < 2|t| < 2 \cdot \frac{\epsilon}{2} = \epsilon.$$

(We used that a larger nonnegative number has a larger square root.) This is exactly the definition of $\lim_{t\to 0}(t, t^2) = (0, 0)$.

Another way (from the review session). We want $|t|\sqrt{t^2 + 1} < \epsilon$. Both sides are nonnegative, so this is the same as squaring both sides: $t^4 + t^2 < \epsilon^2$. A sum of two terms is less than ϵ^2 if each term is less than $\epsilon^2/2$, that is, if

$$t^2 < \frac{\epsilon^2}{2} \quad \text{and} \quad t^4 < \frac{\epsilon^2}{2}, \qquad \text{which hold when} \qquad |t| < \frac{\epsilon}{\sqrt{2}} \quad \text{and} \quad |t| < \sqrt[4]{\frac{\epsilon^2}{2}} = \frac{\sqrt{\epsilon}}{\sqrt[4]{2}}.$$

So $\delta = \min\{\epsilon/\sqrt{2},\ \sqrt{\epsilon}/\sqrt[4]{2}\}$ also works. Both approaches are fine; the first one avoids fourth roots by first agreeing to keep $|t| < 1$.

(b) Give an example of a function $\vec{F} : (-1, 1) \to \mathbb{R}^2$ such that $\lim_{t\to 0}\|\vec{F}(t)\|$ exists but $\lim_{t\to 0}\vec{F}(t)$ does not. Justify both claims.

Solution: The example. Let $\vec{F}$ jump between two points of the same length:

$$\vec{F}(t) = \begin{cases} (1, 0) & t \ge 0, \\ (-1, 0) & t < 0. \end{cases}$$

The length has a limit. For every t, $\|\vec{F}(t)\| = 1$. A function that is constantly 1 has limit 1 (any δ works for any ϵ).

$\vec{F}$ has no limit. Suppose $\lim_{t\to 0}\vec{F}(t) = \vec{L}$ for some vector $\vec{L}$. Take $\epsilon = 1$ in the definition. Then there is a $\delta > 0$ such that $\|\vec{F}(t) - \vec{L}\| < 1$ whenever $0 < |t| < \delta$. Use this for $t = \delta/2$ and for $t = -\delta/2$ (both are in $(-1, 1)$ if we also take $\delta \le 1$):

$$\|(1,0) - \vec{L}\| < 1, \qquad \|(-1,0) - \vec{L}\| < 1.$$

By the triangle inequality,

$$\|(1,0) - (-1,0)\| \le \|(1,0) - \vec{L}\| + \|\vec{L} - (-1,0)\| < 2.$$

But $\|(1,0) - (-1,0)\| = \|(2,0)\| = 2$. This is a contradiction, so $\vec{F}$ has no limit at 0.

The point: the length only remembers how far $\vec{F}(t)$ is from the origin, not in which direction it points. Here the direction flips suddenly at $t = 0$, and the length doesn't notice.

(c) Let $\vec{F}(t) = \left(t^3, \frac{1}{t}\right)$ for $t > 0$. Compute $\vec{F}\,'(t)$ and $\vec{F}\,'(1)$, and use $\vec{F}(1+h) \approx \vec{F}(1) + h\,\vec{F}\,'(1)$ to estimate $\vec{F}(1.1)$. Compare with the true value.

Solution: The derivative. As shown in lecture, a vector-valued function is differentiated one component at a time. The derivative of t^3 is $3t^2$, and the derivative of $\frac{1}{t}$ is $-\frac{1}{t^2}$, so

$$\vec{F}\,'(t) = \left(3t^2,\ -\frac{1}{t^2}\right), \qquad \vec{F}\,'(1) = (3, -1).$$

The estimate. Here $\vec{F}(1) = (1, 1)$ and $h = 0.1$, so

$$\vec{F}(1.1) \approx (1,1) + 0.1\,(3,-1) = (1.3,\ 0.9).$$

The true value.

$$\vec{F}(1.1) = \left(1.1^3,\ \frac{1}{1.1}\right) = (1.331,\ 0.9090\ldots).$$

The estimate is off by 0.031 in the first component and by about 0.009 in the second. Both errors are much smaller than $h = 0.1$: the approximation $\vec{F}(1) + h\vec{F}\,'(1)$ is designed so that the error is small even compared with h.